DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Summer 2022

Course: B. Tech. **Branch : E&TC** **Semester : VII**

Subject Code & Name: BTETPE703B Artificial Intelligence Deep Learning

Max Marks: 60 **Date: 22/08/2022** **Duration: 3.45 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	Explain following approaches of AI i. The cognitive modeling approach ii. The rational agent approach	CO1	**6**
B)	Explain any two foundations of AI.	CO1	**6**
C)	Explain how problem can be defined formally.	CO2	**6**
Q.2	**Solve Any Two of the following.**		
A)	Explain Breadth-first search strategy.	CO2	**6**
B)	Explain the effect of heuristic accuracy on performance.	CO2	**6**
C)	Explain Alpha Beta Pruning.	CO3	**6**
Q. 3	**Solve Any Two of the following.**		
A)	Assume that the underlying physical problem P is defined by $ACTIONS_P$, $RESULT_P$, $GOAL\text{-}TEST_P$, and $STEP\text{-}COST_P$. Explain how we can define the corresponding sensorless problem.	CO3	**6**
B)	Write all three routines of alpha-beta search algorithm.	CO3	**6**
C)	State and explain in brief about all the elements required to define a game as a kind of search problem.	CO2	**6**
Q.4	**Solve Any Two of the following.**		
A)	What is Wumpus World? Precisely define following task environments. i. Performance measure ii. Environment iii. Actuators	CO1	**6**
B)	Write a forward-chaining algorithm for propositional logic.	CO3	**6**
C)	Explain Forward (progression) state-space search.	CO4	**6**
Q. 5	**Solve Any Two of the following.**		
A)	Write GraphPlan algorithm.	CO3	**6**
B)	What is decision tree representation? Explain with suitable example.	CO3	**6**
C)	Explain in brief about Ensemble Learning and Practical Machine Learning.	CO4	**6**

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